



A blockchain-based protocol for tracking user access to shared medical imaging



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ABSTRACT

Modern healthcare systems are complex and regularly share sensitive data among multiple stakeholders, such as doctors, patients, and pharmacists. Patients' data has increased and requires safe methods for its management. Research works related to blockchain, such as MIT MedRec, have strived to draft trustworthy and immutable systems to share data. However, blockchain may be challenging in healthcare scenarios due to issues about privacy and control of data sharing destinations. This paper presents a protocol for tracking shared medical data, which includes images, and controlling the medical data access by multiple conflicting stakeholders. Several efforts rely on blockchain for healthcare, but just a few are concerned about malicious data leakage in blockchain-based healthcare systems. We implement a token mechanism stored in DICOM files and managed by Hyperledger Fabric Blockchain. Our findings and evaluations revealed low chances of a hash collision, such as employing a fitting-resistance birthday attack. Although our solution was devised for healthcare, it can inspire and be easily ported to other blockchain-based application scenarios, such as Ethereum or Hyperledger Besu for business networks.

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1. Introduction

Data protection in healthcare systems has become a critical issue due to the increased confidential and sensitive data leakage. While the amount of data in healthcare systems is at the hype, we argue that data breaches are also growing. Data leakages took place from 2014 to 2019, including the one at Brazilian UNIMED healthcare private corporate [1]. In recent years, data breaches in healthcare providers have resulted from a loss of around \$15 million [1]. The healthcare sector reported an increase of 19.5% of the number of breaches when compared with the ones from sectors [1]. Researchers have been striving to provide a higher degree of protection of personal data through laws to guarantee privacy, such as the General Data Protection Regulation (GDPR) [2].

Computational tools improve medical decision-making by making easy access to data, automating tasks, and suggesting decision-based algorithms. It is also essential to emphasize that

hospitals and research centers have been exchanging information among various collaborators, duly authorized, to assist in diagnosing diseases as part of the research effort [3]. Healthcare systems implement Electronic Health Records (EHR) to register patient reports, exams, diagnoses, and other data types to benefit doctors' management and analysis [4]. A hospital mainly uses medical imaging for reporting exams among the data exchanges, such as head, lung, or shoulder radiography. Medical imaging often relies on a standard format in the medical field, so-called Digital Imaging and Communications in Medicine (DICOM). Such a structure is beneficial as they require common attributes required by the medical systems [5,6].

The healthcare industry is the one most affected by privacy loss. Shi *et al.* (2019) [7] presents a report in 2016 that describes 116 data leaks in healthcare systems. Also, Ismail, Materwala, and Zeadally (2019) [8] state that blockchain technology might reduce risk in sharing health data. Blockchain is a technology that can protect the medical systems and share data, such as patient reports and medical imaging. This technology might preserve security features, ensure immutable logs, provide a trustworthy network tamper-proof-free and auditable [8]. We also argue that blockchain relies on a platform that does not require a trusty third party to manage messages in a decentralized and reliable way through immutable medical records [9].

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Jin et al. (2019) [10] discuss privacy concerns in healthcare and methods to mitigate privacy issues by combining blockchain with other techniques. According to Jin et al. (2019) [10], the cryptographic and anonymization methods have increased the ability of blockchain-based healthcare systems to protect patients' privacy. For instance, to encrypt information, they might use Attribute-Based Encryption and Homomorphic Encryption. Jin et al. (2019) [10] address K-anonymity, L-diversity, and Differential Privacy as anonymization methods. However, this approach views accountability as a technique to follow GDPR rules and guarantee patients' privacy. Our proposal manages shared tokens and audits immutable logs to make the system accountable and tamper-proof-free.

Medical data-sharing solutions fail to address data-leakage auditing by default. A few works exploit auditing approaches for discovering leakages while sharing data, including medical imaging. Patel (2018) [11] proposes a framework that shares medical imaging through a blockchain-based architecture that is free from a trusty third party. However, we argue that this proposed framework is over-simplistic to ((i)) tracking medical shared images and ((ii)) auditing data leaks. The approach still misses following privacy requirements to ensure data leakage accountability. Besides, in conventional systems, the audit log files can maliciously become corrupted as they are mutable.

Medical images accumulate into a high amount of information stored in healthcare systems, and they enable powerful features to be used by machine learning models to extract knowledge [12]. McBee et al. (2018) [6] surveyed the blockchain applications for medical imaging, and they pointed out challenging open issues to be explored by researchers. They claim issues related to privacy, scalability, sharing, and tracking. Consequently, such claims motivated us to focus our efforts on mitigating issues related to medical imaging privacy, tracking, and sharing.

This work sets out a protocol for sharing medical data, including images, based on a blockchain architecture. Our protocol allows for access tracking and a sharing of the medical imaging needed by researchers and practitioners. Unlike other previous research endeavours in this field, our work establishes a protocol for sharing and tracking images reliably while complying with privacy policies and is thus able to hold organizations accountable for leaks coming from ineffective data handling. The benefits of our system are tracking and auditability based on blockchain because if we use a log register and token, they are not immutable. When logs are not immutable, malicious nodes are able to tamper with them.

Unlike previous works, in this study the notion of a unique token is embedded into the DICOM files. By means of auditing, the network administrator can promptly spot the leakages through the logs and unique tokens through this token. In our approach, a token is a hash value embedded in the image metadata and acts as a key to audit leaked images with the support of blockchain.

In order to achieve the goals of this work, we have defined the following hypothesis:

“A blockchain-based protocol can ensure accountability and auditability for healthcare record leakages using immutable logs and a unique token inserted in the metadata of DICOM file images”.

In summary, the main contributions and benefits of this research work are the following:

- A novel protocol that provides traceability and auditability for shared medical imaging;
- A blockchain-based system that ensures that multiple stakeholders with conflicting interests are able to share medical imaging;

- A method of transferring responsibility to those who access images, thus providing data accountability;
- An enhancement of privacy in blockchain-based healthcare systems through robust verifiable data sharing, thus enabling our system to comply with new privacy laws, such as the General Data Protection Regulation (GDPR) [13].

In evaluating our prototype, our results show that the cost for applying blockchain in healthcare system networks is suitable for building in a Cloud environment. We evaluated security analysis, including the tracking token, which proved that the generated hash is hard to collide with using the cryptography birthday attack¹ or by brute force. The theoretical analysis revealed a low probability of tampering with the token by a similar one. Hence, we demonstrated that the blockchain cost of healthcare systems with the tracking token is well suited to meet the new privacy laws, including the GDPR.² Lastly, as proof of concept, we made evaluations with our protocol using radiology DICOM images in the dataset available at The Cancer Imaging Archive (TCIA) from the National Cancer Institute's Clinical Proteomic Tumor Analysis Consortium.³

The remainder of this paper is organized as follows. Section 2 contextualizes the problem that is tackled in this work. Section 3 describes the recent, relevant works in the field. Section 4 introduces the protocol and describes the methods employed to design it. Section 5 reports the experimental analyses and results related to two aspects, namely network performance and security. Section 6 discusses the results and the key contributions of this research. Finally, Section 7 concludes by providing a summary of the work, briefing its contributions, and suggesting future work directions.

2. Problem statement

Across the great diversity of current intelligent and data-oriented systems, especially smart health systems, we could observe there is an urgent need to track time-lined access to patient data. Patient data, such as home phone numbers, are read from digital medical records. Such record accesses are common in Brazil, where some are part of scams; they are then used to make calls asking for cash deposits. Usually, scammers state that they are calling on behalf of the hospital, purposely and wrongly informing that cash is needed for the surgery costs of a family member. Hence, there is an urgent need to track access to patients' data and involved parts, enabling to detect and limiting improper uses.

A distributed privacy-enforcing approach avoids *untrusted centralized control*, which might represent a single point of failure in the whole healthcare system. The tracking is needed as centralized control cannot be fully trusted. A hospital acts as a centralized organization whose staff members might have inappropriate intentions while accessing data.

Governments and institutions are adapting to new standards to enforce privacy more strictly through *privacy laws and regulations*. The General Law for Data Protection (GDPR) has been introduced to guarantee citizens' rights about their data [2,13]. Likewise, healthcare providers have concerns about sharing medical imaging because it contains sensitive information. For instance, DICOM imaging related to X-rays contains metadata that can re-identify patients associated with the exam. Medical images can show parts of one's body before and after the surgery, such as the face and head. Those digital images should not be leaked, even

¹ <https://www.sciencedirect.com/topics/computer-science/birthday-attack>.

² Also, in Brazil, we have the General Personal Data Protection Act (LGPD).

³ Cancer Imaging Archive - [Link](#).

accidentally; thus, we argue that mechanisms that help audit should be devised.

Therefore, the sharing of sensitive data, medical records, must completely support privacy in a decentralized manner where accesses are tracked and verified. The sharing of data can only be consistently implemented through a protocol that enables auditing of accesses to private data, including images. Constraints to accessing and a trace mechanism can hinder data access for malicious purposes while sharing the database among several stakeholders.

3. Background and related works

This section provides an overview of blockchain applied to healthcare systems and research related to blockchain-based systems to track shared assets in healthcare.

3.1. Medical imaging

Electronic healthcare systems use several data types, such as register records, images, and video, to represent patient information. Images are essential since they assist diagnosis and contribute to physicians' decision-making [14]. In information transfer, healthcare providers use a scheme called Digital Images and Communications in Medicine (DICOM) for storing, transmitting, retrieving, processing, and displaying medical imaging information. DICOM protocol can communicate with other physicians or researchers following similar attributes and pixels features [4]. DICOM images have attributes, such as patient identifier, machine model, matrix of slices in grayscale, and others [4].

Research on the healthcare system has investigated medical imaging to study and extract useful information to support the decision-making of the physician [4]. Security concerns in medical imaging need to improve, and approaches are proposed, such as watermarking embedded in images [15]. Watermarking is a technique to improve security to sharing medical images through tamper detection and became image unique [16]. Qasim et al. (2019) [15] ensured both integrity and authenticity of medical imaging through watermarking. The approach inserts a watermark in a smoother Region of Interest (ROI) towards authentication based on image content. It also allows integrity verification of the shared image and protection against image-tampering [15].

Motta et al. (2020) [17] examined a decentralized and distributed information infrastructure following the standard Picture Archive and Communication Systems (PCAS) for sharing medical imaging among healthcare providers in DICOM format. They propose a method called DICOMFlow draws on infrastructure for secure sharing of medical images and improve the system's security. DICOMFlow shares radiological exams through an Internet email structure and defines a global workflow for safe access to images. Different from other approaches, they propose an adaptive infrastructure to manage DICOM images shared using a PCAS system in diverse contexts.

Works in the literature have adopted approaches to improve the security of the images, such as Liao et al. (2020a) [18], Liao et al. (2020b) [19], and Liao et al. (2020c) [20]. Liao et al. (2020a) [18] established a framework for a forensics analysis of the handling of images using a convolutional neural network (CNN). The CNN extracted features that could detect the handling of images and achieved an average rate of accuracy of between 85.96% and 86.12%. Although the CNN achieved a good rate of accuracy, the network has to be trained to obtain features and detect manipulations. Liao et al. (2020b) [19] proposed a steganography strategy based on image texture to determine two payload distributions with the aim of improving security. The framework uses multiple images on the cloud. The steganography image can have adaptive

features depending on the texture of each image. Liao et al. (2020c) [20] employed a steganography method based on color channels to create a new color by including a channel-dependent payload partition strategy to amplify the channel. The aim of this method is to reduce the likelihood of finding embedded content in RGB (Red, Green, and Blue) channels, and it could lead to distortions in textural regions to improve security. Our system focuses on medical imaging and address issues related to traceability and privacy in the blockchain network to improve security, performance and the ability to trace shared images.

This work focuses on medical imaging because of its critical importance in medical diagnosis as a type of media; however, our new approach is designed to adapt to other types of files, such as electronic health records. Medical imaging incorporates valuable features from which knowledge can be obtained through an analysis. Although medical imaging increases the overhead of network resources, this type of media is rich and complex – a DICOM file consists of multiple images as layers. Hence, DICOM databases quickly reach 100 GB storage; when joined to a blockchain network, they have significant overhead that delays the processing of requests. In this paper, a number of issues related to security in medical imaging were investigated and shared for research and advanced science.

3.2. Blockchain in healthcare systems

Blockchain and cryptocurrencies have been used for different applications in the healthcare field since they enable a system to be decentralized, distributed, immutable, free tamper-proof, and smart contracts-oriented (blockchain 3.0) [9]. Blockchain contributes to storing data in a secure shared-ledger, replicated between nodes by a Peer-to-Peer (P2P) network [21], and has led to more reliable systems for data handling. According to Chukwu and Garg (2020) [21] the most used application is the management of electronic health records (EHR), which involves blockchain for healthcare. The authors used a framework, such as Hyperledger Fabric or Ethereum, to build a network – the former is an alternative for constructing a private blockchain network customized and smart contracts-oriented in Golang, JavaScript, or Java. In contrast, Ethereum enables the building of smart contracts based on Ether network [8].

The Hyperledger Fabric architecture, which follows the main components, such as Ordering Service Nodes (OSN), Certificate Authority (CA), world state database, and peers, applies an allowed blockchain. The network's administrator defines limited access for users. OSN is a service that obtains sorted messages and delivers them to nodes. CA authenticates nodes of a network and generates private and public keys. World state database uses Apache CouchDB⁴ for storing records from assets exchanged among nodes. Finally, peers execute and validate transactions associated with the replicated ledger. The architecture also develops smart contracts in different programming languages and customizes assets structures. Among the applications that employ Hyperledger Fabric is our proposal, which sets tools to build a network and smart contracts quickly [22].

Liu et al. (2018) [23] proposed a blockchain network for the sharing of EHRs among healthcare providers using Delegated Proof-of-stake (DPoS) to ensure patient's privacy. The network uses the Proof-of-stake protocol and scheme base-layers, consisted of Data Acquisition, Data Storage, and Data sharing. The authors enhanced privacy through a mechanism called attributed-based encryption for establishing multi-level access. Each attribute applies a different access level Shahnaz, Qamar, and Khalid (2019) [5] described a framework composed of three layers,

⁴ <https://couchdb.apache.org/>.

namely system implementation (smart contracts), blockchain, and user. Off-chain with Interplanetary File Systems (IPFS) ensure secure storage, and a user interface interacts with stakeholders [5].

Although Xuan et al. (2020) [24] did not address healthcare applications, they covered blockchain-based systems for sharing assets. Their work outlined a blockchain model adopting incentives through an evolutionary game theory technique applied to smart contracts. The authors modeled a scheme to describe user participation and find the best way to share data. The strategy increases the amount of shared data for big data applications since it rewards users in the network [24].

Al-Omar et al. (2019) [25] built a blockchain network on Ethereum to control access to electronic health records. Their proposal applies a permissionless protocol by default, called Proof-of-work (PoW), and they add features to protect health data. The primary strategy of this work is to transfer data reliably by using Elliptic Curve Cryptography. Even though they handle cryptographer data, it presents security and privacy issues such as PoW protocol can provide public information that violates privacy rights. Regarding privacy, they address pseudo-anonymization; however, they may suffer linkage attacks and re-identification of patient identity. Besides, they are not focusing on provenance and tracking shared data.

Shen, Guo, and Yang (2019) [26] built a network without using a well-known blockchain platform. They implemented a permissioned protocol based on the Byzantine Fault Tolerance (BFT) strategy. The network is private, where users have defined roles to enter and access data. The handled data are electronic health records containing sensitive patient information, where we needed careful management. The main target of their work is to store health data in the blockchain to control and audit the access. Furthermore, related works do not consider issues about tracking data sharing and privacy. Previous works do not guarantee tracking and patient data leakage accountability; thus, this lack motivates this work to investigate them.

3.3. Blockchain for medical imaging sharing

Section 3.1 describes the way medical imaging in DICOM format can be shared among hospitals, doctors, and research groups for their secure access to data of patients. Blockchain technology enhances the security of healthcare data handling [6]; therefore, applications that use it gain advantages related to confidentiality for the sharing and tracking of medical imaging. In this work, we focused on sharing medical imaging due to powerful features to support decision-making and diagnosis by physicians.

Patel (2018) [11] developed an architecture for medical imaging sharing called images Share Network (ISN), which is based on the network of the Radiological Society of North America (RSNA) for transferring radiological images. ISN aims to design a network with no trust third party, such as a hospital, where the holder of data transfers them directly to the user that requests them. The authors employed a permissionless blockchain through the Proof-of-Stake consensus protocol and improved security using a public and private key scheme to access the images [6,11].

This work address issues on sharing medical imaging due to powerful features of images to support decision-making and diagnosis by physicians [12]. Artificial intelligence leads to high success in solving images issues. Medical imaging can utilize tools to improve diagnosis. However, artificial intelligence lacks access to data for healthcare research [27]. Our solution aims to share medical imaging based on blockchain to provide safe and reliable data holders, thus sharing more data.

3.4. Remarks

Table 1 shows studies related to blockchain in healthcare, and most of them apply a proof-of-concept on EHR and build a customized blockchain architecture to solve a specific issue. They also employ a consensus protocol based on PoS since they are lighter than PoW, speed the validation process, and use a public blockchain network. Nevertheless, the construction of prototypes based on public networks is not the best strategy since anyone can access the application, which is against security requirements of healthcare systems [9]. In similar studies, privacy requirements address only a crypto method for hiding messages; no accountability method is applied, and no privacy regulation is followed.

In contrast, our research approaches DICOM image sharing, as in [11] but adding provenance, accountability, tracking, and policy privacy. Our protocol's main features are associated with tracking, provenance, and accountability, thus meeting the requirements of LGPD privacy policies. The chief ones to be covered are lawfulness and transparency so that the owner is provided with the way data are handled [28]. The protocol based on a proof of concept system also enables analyses of the provenance of leaked data through the reference token stored in the blockchain.

This work also describes a method that uses a token scheme for tracking DICOM images shared among researchers, who can analyze the provenance of data for detecting the organization responsible for poorly handled data. Our protocol enables holders to share data so that they can collaborate with other researchers securely. In other words, it ensures tracking and provenance in linked data so that network administrators can adapt their systems to privacy regulations.

4. Blockchain system and protocol to trace medical images shared

This section describes the components of our solution, such as blockchain network, protocol design, protocol flow, and the method for the token generation. Fig. 1 displays an overview of the proof-of-concept (POC) system that our protocol applied. The roles in the figure display researchers and doctors as requesters and hospitals as providers, in which the patient is the data holder, interacting with the hospital to make exams.

4.1. Role players and system protocol

The system design depicted in Fig. 1 provides a big picture of role players in tracking and auditing the sharing of sensitive medical data. The Proof-of-Concept (PoC) comprises a blockchain network, a healthcare provider, a tracking token mechanism, a query application interface (API), and an arbitrary application towards ensuring security. Besides, the protocol covers main algorithms for sharing medical imaging studies, requests of the studies, and audit logs.

Researchers, doctors, and healthcare providers perform critical roles in the envisioned data sharing in the proposed design since they may potentially generate data leakages. In short, the devised tracking and auditing system contemplates the following roles:

- **Holders.** They are data owners, such as the patient who provides images for study.
- **Provider.** It is a healthcare organization that handles patients' data, such as hospitals or research centers.
- **Requesters.** They are people or organizations that need data for studies. They request data access to a provider that forwards the request to the holder.

Table 1
Summary of related works.

Work	Features		Blockchain platform	Consensus algorithm	Network type	Provenance	Tracking	Privacy policy
[23]	EHR	No	Unknown	DPoS	Unknown	No	No	Yes
[5]	EHR	IPFS	Ethereum	PoW	Permissionless	No	No	No
[24]	Anyone	No	Unknown	Own	Own	No	No	No
[25]	EHR	No	Ethereum	PoW	Permissionless	No	No	No
[26]	EHR	No	Unknown	BFT	Permissioned	No	No	No
[11]	DICOM	No	Unknown	PoS	Own	No	No	Yes
This	DICOM	No	Hyperledger fabric	PBFT	Permissioned	Yes	Yes	Yes

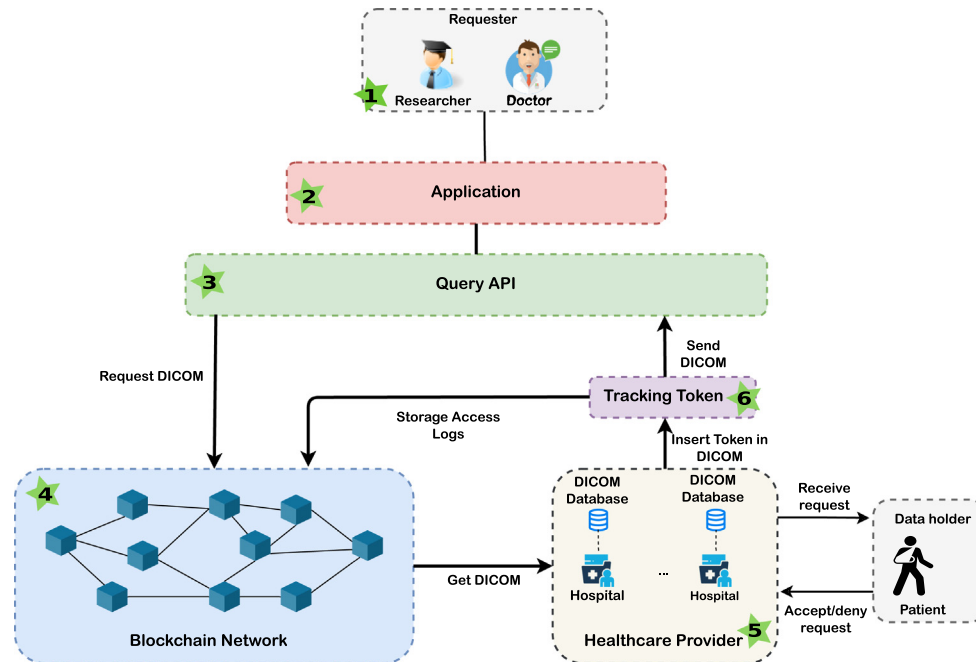


Fig. 1. Proof of concept of the system designed for the tracking and auditing of shared DICOM images from a unique token.

- **Researchers.** They request data from providers and holders to conduct research work and studies. They collect findings to advance science and a research field. Besides, they generally are joined to universities or institutes, and they can be doctors.

The main component of our proposal comprehends a protocol for sharing medical imaging through a blockchain structure to store the images files, references, and audit logs. The DICOM files are stored in a server using the SSH File Transfer Protocol (SFTP)⁵ to access and recover files. The protocol is composed of the following algorithms. (i) Algorithm 1 comprise the functionalities that allow researchers and practitioners to request data for studies. (ii) Algorithm 2 enables the data holder tasks to check and accept a request to access its images stored on the provider. (iii) Algorithm 3 introduces and manages audit logs through hash tokens stored into medical imaging; it also assists the network admin to identify data requesters.

In Algorithm 1, researchers or doctors send requests to the healthcare provider that forward to the data holder to accept or decline to share its medical imaging. Each requester S_i can request several images of Im that they can use in research (line 2). Before proceeding with a request, the blockchain network verifies whether the requester is valid, and thus it is enabled to make a call to the system (lines 3–4). We assume that the Hyperledger Fabric framework allows organizations to ingress the network

through its authentication method by Cryptography of Elliptic Curves. The requester sends a message to healthcare providers, and waiting for a positive or negative response (lines 6–9). If the data holder accepts the request, the stakeholder can access and query their medical imaging (lines 8–9). The following algorithms complement the protocol design, and the algorithm comprehends the whole protocol for sharing medical imaging.

Algorithm 2 aims to share images data from holder to researcher or doctor, so patients wait for requests. When it receives a request $rc \in RC$, the system extracts the message header and obtains the image identifier (lines 2–3). Then, the system collects the image id from request aim to get images from the Healthcare provider repository (line 4). The system verifies if the search returned valid images (lines 5–6), and it inserts a different hash token into each medical images' metadata from the search results (line 8). The system creates the token based on a 256-bit hash value that is obtained from joining the respective patient identifier and access timestamps (line 7). The timestamps follow RFC3339,⁶ which considers the system's local time to generate a large number. Finally, the system generates access logs to identify requesters (line 9). These logs are stored in the blockchain ledger because they are immutable, accessible, and free of tampering (line 10). Previous works have employed different methods to prevent tampering with sensitive data. Qasim et al. (2019) [15] presented a method to protect images by watermarking through

⁵ <https://datatracker.ietf.org/doc/html/rfc913>.

⁶ <https://datatracker.ietf.org/doc/html/rfc3339>.

Algorithm 1: Image Request

Data: R : requester set;
 BC : blockchain;
 SH : stakeholder set
Result: RL : request list

```

1  $RL \leftarrow \emptyset$ ;
2 foreach  $r \in R$  do
3   if  $r \ni BC$  then
4      $\text{return } RL$ ; //  $r$  invalid
5   foreach  $im \in IM$  and  $im \in s_{IM}$  do
6      $S \leftarrow \text{request}(im, SH)$ ;
7     foreach  $s \in S$  do
8       if  $s == \text{accept}$  then
9          $RL \cup im$ ;
10        break;
11 return  $RL$ ;

```

Algorithm 2: Image Share

Data: RC : receive images set;
 BC : blockchain;
 HP : healthcare provider
 P : patient
Result: IML : image request list

```

1  $IML \leftarrow \emptyset$ ;
2 foreach  $rc \in RC$  do
3    $im_{id} \leftarrow rc.image.id$ ;
4    $im \leftarrow \text{search}(im_{id}, HP)$ ;
5   if  $im \ni HP$  then
6     continue; // no  $im$ 
7    $token \leftarrow sha256(P.id + \text{Timestamp}_{access})$ ;
8    $im.metadata \leftarrow \text{write}(token)$ ;
9    $log \leftarrow \text{generate}(time, P.id, im.id, token)$ ;
10   $BC \leftarrow \text{createLog}(log)$ ;
11   $IML \cup im$ ;
12 return  $IML$ ;

```

Message Digest 5 (MD5) algorithm. MD5 is faster than SHA-256 to generate a watermarking because it makes use of 128 bits [29]. However, MD5 has several known vulnerabilities, such as no collision resistance (#CVE-2004-2761)⁷; thus, our method presents collision resistance and more effectively enforces more security for patient's images.

Algorithm 3 performs the auditing of logs by the auditing entity SH using the token as the primary key to recovering accesses over the images, including leak images. First, the log-auditing algorithm verifies if the current entity's role, the role of the stakeholder, on the blockchain is as *admin* (line 1–2). Then, stakeholder SH uploads the UL that corresponds to leaked imaging (line 5), and the system collects the access token from the image (line 7) to identify which requester accessed the image using the blockchain logs. The hash tokens in a *token* list are the key to search logs that match them (line 8). At last, if the search identifies existing matching logs (line 9–10), the algorithm returns a list with found attributed-value JSON *log* values (line 11–12). These values can attest the requesters who leaked the medical imaging. The auditing entity A then knows about the token to query data on blockchain about the following pieces of data: (i) where is the token related to requester; (ii) one database row on

Algorithm 3: Leaked Image Auditing

Data: UL : upload leaked images set;
 BC : blockchain;
 SH : stakeholder
Result: L : logs list

```

1 if  $SH \ni BC$  and  $\text{role}(SH) \neq \text{"admin"}$  then
2    $\text{Return Error}$ ; //  $SH$  invalid
3  $L \leftarrow \emptyset$ ;
4 foreach  $ul \in UL$  do
5    $IM \leftarrow ul$ ;
6   foreach  $im \in IM$  do
7      $token \leftarrow \text{extractHashToken}(im)$ ;
8      $log \leftarrow \text{searchLogByTokenId}(token)$ ;
9     if  $log == \emptyset$  then
10       $\text{continue Error}$ ; //  $log$  is empty
11      $L \cup log$ 
12 return  $L$ ;

```

the blockchain has requester ID; (iii) the user's identity to check who request images; and (iv) reference token about the request.

4.2. System design

This subsection describes the abstraction of the system proposed and how it was designed. We present all modules developed, such as protocol to enhance security, the blockchain network, and trace mechanism.

The proposed protocols enable to enhance security and privacy in healthcare environments, and their functionalities are characterized through a bottom-up design within an electronic health information exchange system. This system demonstrates the efficacy of enforcing data sharing through methods and models based on blockchain (see Fig. 1). Our PoC and protocol were designed to address privacy concerns on the basis of access to auditable and provenance features. There are papers in the literature such as *Maesa, Mori, and Ricci (2019) [30]*, *Marbough et al. (2020) [31]* that describe provenance, tracking, and auditability as benefits for blockchain-based systems. In healthcare settings, privacy (followed by auditability) can be an essential feature, and our system seeks to address this. Finally, according to *Marbough et al. (2020) [31]* privacy in blockchain represents a system where selective information is shared; thus, our system shares medical imaging by granting access to selected users and tracking assets shared by the unique token.

The first module (Module 1 requester, gray): this has an interface for communication among its entities (entities are roles such as a healthcare provider, requester, or administrator), blockchain network, and healthcare provider (Module 2, light pink). The entity requests images from the API query (Module 3, green) that have to be sent to the healthcare provider, and the system registers the requester access as a log in the blockchain. When the data holder decides to make an image available, the access log is stored, and the image is sent through the token generator procedure. The asset remains available on the API query interface so that it can be accessed by the application stakeholders, such as doctors, patients, and researchers.

The blockchain-based network (Module 4 in blue color) stores data towards a reliable and immutable system. The immutability of blockchain allows that data stored is not being removed; if the system writes data in the ledger, they are hard to erase, and it helps to improve accountability of the system. Access logs and images' metadata guarantee security to the system to control

⁷ <https://nvd.nist.gov/vuln/detail/CVE-2004-2761>.

tamper-proof and audit logs [32]. A smart contract implementation was defined for auditing the logs stored in the blockchain and comprises the following attributes: token identification, timestamp access, and access level. It also includes the stakeholders and organization that sent images and the stakeholders and organization that requested them. The attributes defined in smart contracts are stored in the blockchain towards helping the admin network identify leakage. The smart contract establishes rules for transactions and receives assets, such as DICOM images [33]. The network was built in a Hyperledger Fabric structure with 10 Dockers peers, two organizations, two certificate authorities(CA) for each organization, and one orderer.

The module (Module 5, yellow) healthcare provider is an image repository from healthcare organizations. It comprises hospital systems of image databases, which register file references in a blockchain network so that they can be shared with other entities within a healthcare research network. The hospital uses a connector to register images in the blockchain to make them available for researchers and doctors.

The tracking token mechanism is the system's main feature. It creates a hash value that, together with metadata, is inserted in DICOM. This value, called "token", is associated with the log structure, such as the stakeholder's identifier, and stored in the immutable ledger in the blockchain. The following format was defined from algorithm SHA-256 with 256-bit message size digest and 32-bit word [29]: $token = sha256("PATIENT_{ID}" + "TIMESTAMP_{ACCESS}")$. The token design is a hash value with a unique property that can produce traceable images in case of leakage or improper management. After the token generation and storage in the blockchain, the images are sent to the API interface to query results.

The blockchain network aims to store immutable logs and check image leakage, which can be audited by the token inserted in DICOM and metadata. Only a single stakeholder can access the image. Therefore, organizations can detect the entity or stakeholder who has made the image available, analyze the data provenance, and, by following privacy regulations, decide on actions to be taken.

The last module incorporates the application and stakeholders (doctors, researchers, and healthcare providers). The module is designed to enable the developer to create an adaptive system for healthcare image sharing as a friendly interface (Module 6, purple).

Ideally, any other developer using the proposed system can integrate its application seamlessly. Stakeholders use the application to obtain assets. At this point, the access level is defined with researchers and doctors through the reading of permissions, where healthcare providers read and write the permission document. On the other hand, the healthcare provider can be a hospital that owns the data. The researcher is an individual or an organization dedicated to the extraction of knowledge on health data.

4.3. Protocol in the proposed system

Fig. 2 shows the flow of the protocol together to algorithms for requesting images, sharing images, and auditing logs through the blockchain structure, as well as healthcare provider medical imaging repository.

In the sequence diagram built based on Unified Modeling Language (UML), the doctors, researchers, and healthcare providers also are called stakeholders. Stakeholders query data from an interface that connects to the control line query API for requesting images. **Query API** interprets events that occurred in object $\llcorner$: *interface* $\gg$ and relays them to the Blockchain, responsible for data management, storage, devices, message broker, flow,

logic apps, logs, distributed ledger technology, and cryptography hash. The POC implementation of our system design comprises a Python⁸ interface for clients to communicate with the blockchain network and connect to a query API in Nodejs.⁹

Fig. 2 shows the process before a request is sent to obtain DICOM images. The Certificate Authority of blockchain checks whether the stakeholder (any role of network) is valid, and they can join the blockchain. After validating the stakeholder, the system can determine their identification and control their access level to the assets. **Query API** requests image data from the Blockchain, which also obtains them, but from a healthcare provider. In our POC implementation, we select DICOM images from a public dataset.¹⁰ The requester (doctor or researcher) that interacts with the system can demand images DICOM, which were previously downloaded and stored in the healthcare provider of our POC system. A token is then inserted in each DICOM file when a requester demand an image and the request is allowed. Next, the system sends images to the healthcare provider (where images are in its database), which uses a service based on the SFTP to send the images to the query API and make them available in the interface application.

The workflow implemented adopts a smart contract in Golang,¹¹ the blockchain network configured with Hyperledger Fabric¹² in the Docker containers. In the setup of the Fabric network, a bash script serves as the interface for customized access to peers, channels, organizations, and orderers. Ordering nodes are responsible for managing and ordering transactions of organizations that can be part of the blockchain. Our POC implementation comprises a stakeholder interface and a module healthcare provider service, which sends files and connects to logs stored in the Blockchain. The logs are sent as a JSON file and help audit images leakage using the token that links to a blockchain stakeholder.

The source code of our POC implementation is available at open repository **blockchain token Dicom**.¹³ This code enables researchers to replicate experiments and extend the system and protocol. It splits as a Hyperledger network representing the blockchain configuration with docker files, certificates, and bash scripts. Besides other folders, such as an API for direct communication with blockchain, the shared Dicom folder contains the python scripts *Server* and *Client* to exchange files between stakeholders and blockchain.

Finally, Hyperledger Fabric architecture manage keys, storing, sharing, and checking stakeholders' credentials to access medical imaging. According to Fabric's documentations,¹⁴ elliptic curves cryptography, based on algorithms of curve prime256v1¹⁵ and SHA256, are employed to generate public and private keys aimed at stakeholder joining the network. Each stakeholder on the network has a role, and they can be providers, requesters, or data holders. Thus, the blockchain creates digital certificates for user login to a hospital or its wallet. All credentials are management to a blockchain Certificate Authority (CA). Thus, the peer CA manages credentials and stores on blockchain if entries are valid.

⁸ <https://www.python.org/>.

⁹ <https://nodejs.org/en/>.

¹⁰ Cancer Imaging Archive - [Link](#).

¹¹ <https://golang.org/>.

¹² <https://github.com/hyperledger/fabric>.

¹³ Github - <https://github.com/eriksonjAguiar/Blockchain-Token-DICOM>.

¹⁴ <https://hyperledger-fabric.readthedocs.io/en/release-2.2/identity/identity.html>.

¹⁵ <https://datatracker.ietf.org/doc/html/rfc8422>.

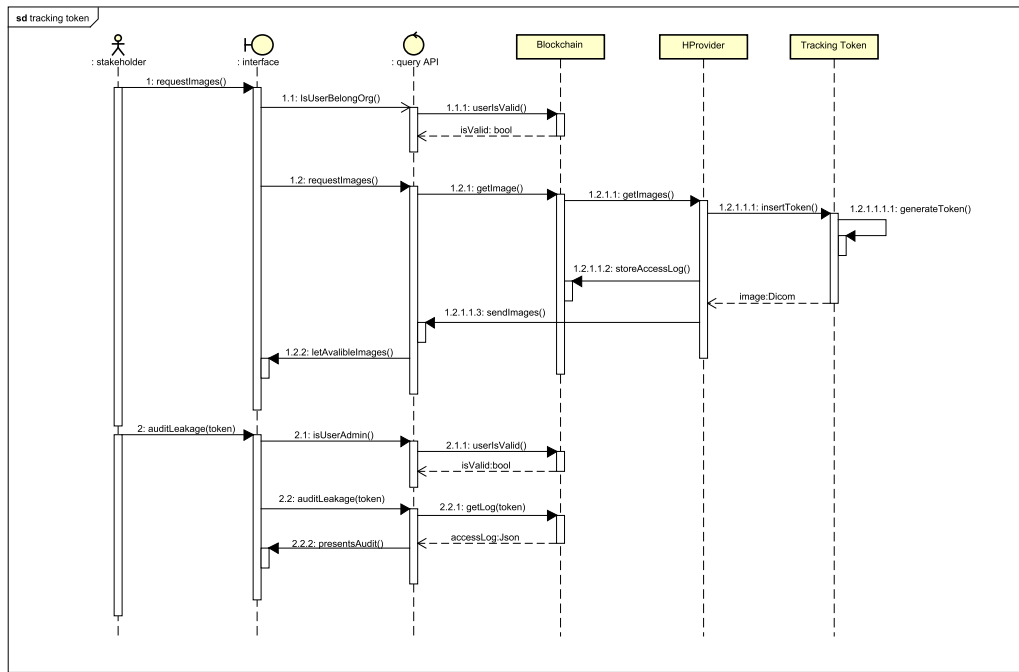


Fig. 2. Sequence diagram describing the proposed protocol (Algorithms 1, 2, and 3): requesting, sharing, and auditing assets in the blockchain.

5. Experiments and results

We have conducted experimental analyses to demonstrate and validate our proposal using the Hyperledger Fabric platform. The platform has been set up over a Python script regarding two aspects, namely blockchain network performance and security requirements quality. The evaluated security requirements followed the quality of a token-generating script based on Python.

5.1. Experimental setup

The experimental platform consisted of the Blockchain framework Hyperledger Fabric running on 10 Docker containers, which represented the full nodes of the network for the performance evaluation. The server's hardware comprehends an AMD Ryzen 7 CPU of 32 GB Ram and Solid-state drive (SSD) for high-speed data persistence in the blockchain world state database. The database used in the experiments had around 7.5 GB of DICOM files from the National Cancer Institute's dataset [34,35]. We sent the images from the database emulated locally, resulting in low latency to query. However, when the dataset joins with blockchain, the latency grows. We ran experiments 30 times, considering results with a 95% confidence interval. We conducted our experiments in a single machine by simulating a distributed environment composed of peers, healthcare providers, patients, and requesters (doctors and researchers). Likewise, we employed the Hyperledger Caliper to organize simulations and configure nodes, distributed machines, on the network.

Furthermore, we configure a single machine with Hyperledger fabric blockchain, our API, providers, and requesters to execute these algorithms. Although we used a single machine, we built a simulation environment for distributed systems from Hyperledger Caliper.¹⁶ The environment was composed of blockchain peers, healthcare providers that store images, and requesters who want to get imaging to study.

We set up our experiments in a server Linux, where the POC system process the transactions is separate in Docker's containers to simulate the valuable components. We use the TCIA dataset accessed from the SSD of the local server to create an image repository. The image repository was configured as a container simulated in a local server to communicate between other containers such as Healthcare providers, Data holders, and Requester. In simulations, we built ten nodes as dockers containers to compose a blockchain network, and each stakeholder node communicates with Certificate Authority, Database, Orderer, and API to register transactions. We simulate all nodes in a decentralized network, with nodes communicating without a third party to manage transactions and share images. Finally, our simulations have actors nodes, such as Requester, Healthcare Providers, and Data holder. A Requester demands an image from a Healthcare provider that stores images in a repository and the Data holder that allows/declines access to its data. The healthcare provider is a container that communicates with the data holder, requests access to data, and forwards images to the Requester after the owner is allowed. In summary, we simulate the blockchain network and collect metrics using the Hyperledger Caliper.

The security features validate the hypothesis based on the evaluation of hash security quality. The properties of the token were evaluated by Levenstein's similarity and Shannon entropy [36]. Token properties are essential to validate that the hash is collision-resistant and can be a component to help the system to be tamper-proof-free.

5.2. Network performance analysis

The analysis of the blockchain network's performance aimed at ensuring a low cost of Hyperledger Fabric network for the exchange of assets, differently from Bitcoin and Ethereum. Figs. 3(a) and 3(b) show the growth in CPU and memory usage, respectively. Regarding the testing execution, a significant amount of transactions were submitted to transfer images between network peers.

The network latency was checked, as displayed in Fig. 3(c), towards faster and more suitable transactions for healthcare systems. The boxplot related to each network part illustrates (i)

¹⁶ <https://github.com/hyperledger/caliper>.

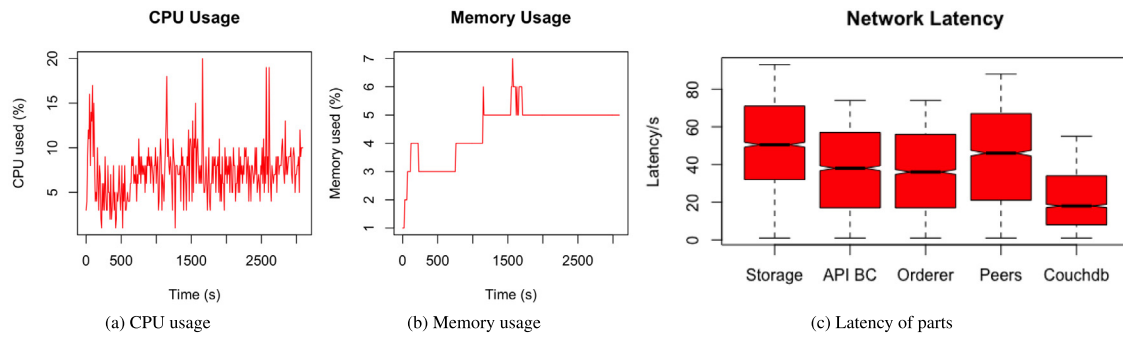


Fig. 3. Performance analyses of the proposed proof of concept in terms of consumption of system resources.

Table 2

Performance analysis of components in the blockchain network.

	Metric	Mean	Median	Standard deviation	Maximum	Minimum
Hardware usage	CPU %	7.33	7	2.94	20	1
	Memory (%)	4.43	5	0.96	7	1
Network latency	Storage (s)	49.96	50.50	24.84	93	1
	API BC (s)	37.34	38	21.72	74	1
	Orderer (s)	36.77	36	21.55	74	1
	Peers (s)	45.45	46	25.21	88	1
	CouchDB (s)	20.22	18	13.52	55	1

latency in image files transferred from the healthcare provider to the requester, (ii) latency in interaction with blockchain external user, (iii) latency of the orderer (orderer are nodes responsible for managing and ordering transactions of organizations) that communicates with all peers, (iv) latency in peers interaction with the smart contract installed, and (v) latency of CouchDB Hyperledger Fabric database. The blockchain performance evaluation represents the exchange of information in a healthcare scenario. For such a use case, the network shows increasing peaks of memory and CPU. Our hardware evaluations consisted of receive transactions in blockchain from requesters to healthcare providers. The communication interval outlined for a distinct part of the network identified the highest latency in connection with the image files stored and peers since storage has sizeable amounts of DICOM files and peers interact directly with blockchain, thus causing delays.

Regarding overhead from statistical metrics, we measured a central position and dispersion of the values found. Table 2 lists such metrics for a summary of samples, describing the characteristics of the variables. The hardware metrics display low machine resource usage, resulting in average values of 7% CPU usage and 5% memory usage. The hardware quantities analyzed revealed a low variance of CPU and memory. Therefore, their values are close to the expected one – the mean value. The results also showed that the hardware is suitable for applications to medical systems since it consumes limited resources of the machines, and it is scalable for a large number of peers using the Hyperledger Fabric framework. [22]. The network evaluation considered the latency values for each part of the network. Each part of the network required a longer time, around 50.50seconds on average, for sending assets to storage in the blockchain due to the large size of the DICOM files. The blockchain network also showed high latency due to consensus mechanisms and the necessity of node response for validating transactions. However, Hyperledger Fabric blockchain is light for sharing medical imaging compared to other alternatives, such as Bitcoin and Ethereum [22].

5.3. Security performance analysis

We checked the quality of our proposal's security requirements from the similarity and entropy of the generated hash

values, namely tokens. It trailed the *birthday attack model* to check possible hash collisions from similarities in the hash found in DICOM files [37]. The *birthday paradox* enabled the probability of users of the same birthday date being in the room. The method can also map the hash collisions probability using bit numbers, such as in SHA-256 algorithm, for the counting of collision values [38].

We quantify the similarity among generated tokens, aiming to analyze their properties by calculating the Levenshtein distance of each token. This process represents the probability of generation of a new hash token similar to the one contained in the DICOM file [36]. The token describes a key for auditing stakeholder access to DICOM images. Levenshtein distance defines closer to zero, more similar two values are; therefore, a token is close to another new one in case of a small Levenshtein distance. Malicious users might try to generate equal hash values for tampering access and performing image leakage. Shannon's entropy, given by Eq. (1), where each value for $x_1, x_2, \dots, x_n \in$ a random variable X , compares the randomness level of the token generated. Therefore, collision attacks by brute-forcing hash values are non-feasible to tamper since the probability of an attacker inferring with the actual hash from a distribution pattern is very low [39].

$$E_t(X) = - \sum_{x \in X} p(x) \log_2 p(x). \quad (1)$$

Our solution requirements were validated to combine blockchain for auditing and token hash to create unique DICOM images. The validation step analyzed the probability of the system resisting attacks for hash invalidation or collision through experiments.

The first set of analyses investigated entropy and similarity regarding token quality, as summarized in Table 3. Such quantities are essential for studies on typical and variable values, obtaining a string similar to one of the tokens was generated in previous iterations.

Table 3 shows the entropy and similarity values to describe the low probability of the malicious node getting a similar tracking token, which aims to tamper with the network. Thus, a low probability increases the complexity of finding a valid token for tampering. This difficulty in tampering allows for security and

Table 3
Statistical analysis of security aspects.

Metrics	Security values	
	Similarity	Entropy
Mean	52.36	3.80
Median	55	3.81
Standard deviation	12.17	0.04
Maximum	60	3.88
Minimum	50	3.70

privacy even if data leaks and a malicious node invalidates its accountability to leakage; the high computational efforts make tampering infeasible.

The similarity outcome with 53.36 mean value showed a long distance between the hash tokens generated and a new random token. The characterization of tokens should express distinct patterns – the obtaining of identical values by a malicious node is challenging. Besides, the dispersion metric for similarity quantity is high since its standard deviation is 12.17 and represents scattered values that prove the randomization of those generated and distributed in the sample space far from the mean.

Entropy represents an uncertainty metric for discovering an attribute's approximate value, and an attacker cannot be sure of the estimate of the actual value [39]. Towards entropy results, we obtained a 3.80 mean. At this point, a 0.04 standard deviation showed a low probability of an attacker discovering a similar token and causing conflicts to the network. Entropy also shows low variability due to low standard deviation, representing most values close to each other. Suppose malicious nodes know the system's characteristics. In that case, they might attempt to determine a value through brute force following known features. Similarly, such attempts resemble the launching of a coin with 2256 sides and the expectation of its falling on the side chosen by the user attacked.

According to the literature, another method to measure the probability of hash collisions consists of using Eqs. (2) and (3) to calculate the amount of messages sent, N_m , where BW is the bandwidth, MZ is the message size, and T is the time spent on the sending [40]. Eq. (3), where n denotes the message bits number and N_m is the amount of messages sent, calculates the probability of collision P [40]. We collected packets on network by using Wireshark tool,¹⁷ which allowed us to obtain the following values: $BW = 9.2$ GB, $MZ = 671$ bytes, $T = 10$ s, and $n = 256$.

$$N_m = \frac{NW}{MZ} \times T$$

$$= \frac{9.2 \times 2^{30}}{671} \times 10 = 147,219,445 \text{ messages} \quad (2)$$

$$P = \frac{N_m}{2^n}$$

$$= \frac{147,219,445}{2^{256}} = 1.27 \times 10^{-69}\% \quad (3)$$

The experiments resulted in a low probability of hash collision – the method used was proposed by *Al-Ani et al. (2019)* [40] with a $1.27 \times 10^{-69}\%$ chance of finding a hash token of similar value in a complete set of tokens. For instance, a brute force attack hardly obtains collision values since it requires several attempts and high computational costs to be effective.

Probability Density Function (PDF), which describes a parameter estimator that approximates the actual values for the entropy, explored the behavior of entropy variables. To plot the curves, the values were normalized from 0 to 1 – [0, 1] interval. Fig. 4(a)

shows the entropy peaks of higher concentrations between 0.30 and 0.35 and values spreading with small 0.25 concentrations. The distribution shows a low probability of a parameter estimator obtaining a value from 0.20 to 0.25 for entropy; therefore, the entropy behavior means the values have a higher probability of converging towards a 0.35 entropy quantity. On the other hand, it indicates that a malicious node has a low chance of learning a token pattern for attacking the system due to higher entropy values.

Cumulative Distribution Function (CDF), illustrated in Fig. 4(b), completed the entropy quantity analysis. We ran for theoretical curves in normal, Gamma, and chi-square distributions compared to empirical CDF for a normalized interval of entropy values [0, 1]. Gamma is the farthest from entropy – red line, and chi-square and normal distributions exhibited the most similar behavior among the distributions. Chi-square – green line – best fitted the actual values, thus implying an asymmetric distribution. The randomness value of the token hampered the obtaining of a similar token value.

The last examined component is the similarity normalized to [0, 1], representing Levenstein distance, which describes the behavior distance between hash tokens. Most distances represent several characteristics different from actual hash values. The PDF plotted in Fig. 4(c) for the analysis of component similarity illustrates a role with peaks of maximum values. Therefore, hash token values have a considerable distance and low probability for obtaining a hash with similar aspects.

CDFs comprehend the similarity and entropy variables according to normal, gamma, and chi-square distributions. Fig. 4(d) shows the CDF behavior for similarity. The behavior of normal distribution – blue line – is comparable with that of similarity – red line, and Gamma distribution – orange line – has a similar behavior. Normal distribution fits natural distributions of the population around the mean and our findings of similarity is compatible with distribution's population [41].

6. Discussion and limitations

In the following, we discuss the key aspects and significant contributions for each feature highlighted below:

- **Data Sharing.** Our proposed protocol enables the sharing of medical imaging among researchers, supporting secure and private exchanges for exams and helping diagnosis of diseases;
- **Blockchain platform.** With the blockchain platform, the system might be adaptive for others approaches and offer a reliable tool to build a blockchain network for the healthcare environment;
- **Consensus algorithm.** We used the blockchain platform Hyperledger Fabric that uses a light consensus protocol, the Practical Byzantine Fault Tolerance (PBFT). Thus, it takes less time to validate a transaction and generate less overhead on the network than other blockchain platforms;
- **Network type.** Considering the healthcare environment handle sensitive data, we use a private blockchain network because reliable peers control it and only specifics nodes can validate a transaction;
- **Provenance and Accountability.** The auditing entity can use the trace token to provides the researcher knowledge about the source of the data, and it is a safe way for researchers to acquire data. Besides, data requesters can be accountable if data is leaked.
- **Tracking.** Our protocol addresses a tracking token stored in a blockchain that is hard to tamper with due to hash randomness. Moreover, our results prove it is hard to find a similar token through a birthday or brute force attack to tamper who leaked the data;

¹⁷ <https://www.wireshark.org/>.

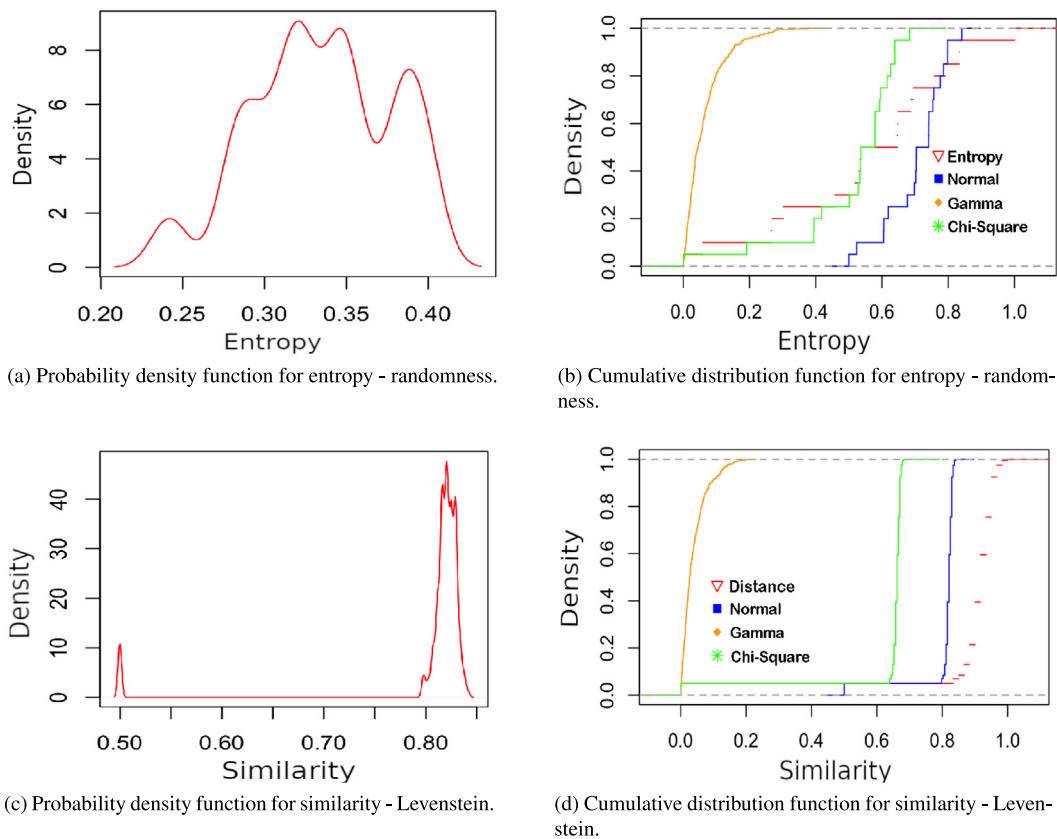


Fig. 4. Statistical distributions for hash quantity values.

- **Privacy policy.** Our approach provides a more reliable system for complying with privacy regulations; it addresses the question of accountability with regard to data leakage detection when the sharing is poorly handled. This process is based on immutable audit blockchain to find the stakeholder (who has a requester role) who requested the leakage image through a token stored in the metadata.

According to our findings, we believe the blockchain-based protocol for tracking shared DICOM images can contribute to auditing, provenance, and protection of healthcare systems and provide accountability of stakeholder requests for sharing data. Moreover, it has responded to our following hypothesis:

“A protocol blockchain-base can ensure accountability and auditability for healthcare record leakages using immutable logs and a unique token inserted in metadata of DICOM file images”.

For characterizing the protocol to sharing medical imaging, our research has revealed that an attacker has a low probability of discovering a similar token from patterns known by the network. Therefore, the analyses metrics demonstrated that the randomness feature of the token hampers collision or birthday attacks for tampering with the token in DICOM images.

The protocol design ensures immutable logs for auditing tamper-free by the Information Technology Manager. The auditing entity can analyze the provenance of data employing the blockchain structure, and data holders can consult the purpose of data use. The hardware metrics proved the system's suitability for applications to healthcare systems according to its acceptable latency values, throughput, CPU usage, and memory usage. The system can be added to usual healthcare systems for enhancing their security and reliability. Thus, researchers can access traceable data for protection against leakages.

The experimental results clearly answered our hypothesis, revealing that the applied system has large entropy. Similarity implied a tracking token of hard breakage, since an attacker must have a more profound knowledge of the network for inferring a valid token. Besides, the system offers an interface for auditing logs by the auditing entity to identify DICOM imaging leakages and transfer leakage accountability to the stakeholder that requested the images. Therefore, stakeholders obtain images based on logs stored in the blockchain and use the token as the recovery key.

Experimental results also allowed to characterize the protocol's behavior in sharing medical images based on PDF and CDF analyses. Even though it shows a token, the behavior appears to have a low probability of a malicious user getting a similar value. The token can enhance the reliability to distribute the images, and it guarantees who access it.

In terms of this work's limitations, we can mention that we did not implement an interactive interface for entity to audit logs and analyze provenance. Furthermore, only one image format was adopted in our study – DICOM. Nevertheless, we argue that other media formats can be easily included as our implementation is highly modularized. We addressed only one type of attack, and our proposed approach can be tested against other types of attacks, such as Sybil, Distributed denial-of-service, Routing attack, and Eclipse attack [42].

7. Conclusions and future work

This research article has outlined the problem statement and presented a protocol for tracking requesters' access to shared medical data, including images. We have conducted several studies on the existing gap and identified that tracking medical imaging has not been fully addressed. Hence, our work targeted this

kind of data in our proposed system. This entailed investigating how blockchain technology could be used as a tool both to ensure and enhance security and privacy for data sharing. On the basis of well-established frameworks, we implemented our prototype in Hyperledger Fabric. This platform enabled us to work out a solution based on a lightweight blockchain consensus protocol suitable for healthcare systems interfaces.

The network overhead and performance revealed similar figures towards a centralized network in terms of throughput, CPU, and memory. The proposed method and protocol demonstrated a low probability of a malicious node obtaining an equal token. These findings show that our proposal helps in protecting against hash collisions that break the token's security.

Also, similarity and entropy evaluations revealed that the stakeholder's brute force attacks are unfeasible for tampering or invalidating the tracking token. We highlight that the hash token can provide accountability and give a more secure environment for patients to share data. Health data access was increasingly higher during the pandemic year of 2020, in which several COVID patients checked in to hospitals worldwide [43].

As future work, we aim to develop an off-chain based on a P2P network that provides distributed privacy and mitigates attacks to image databases. This work can then meet more requirements related to privacy regulations, such as Health Insurance Portability and Accountability Act (HIPAA). Besides, our protocol might adapt to the Ethereum network because we build from Hyperledger tools, and we can interact with Hyperledger Besu¹⁸ to build a business network based on Proof-of-work and Proof-of-Authority protocols, running our protocol to track shared data.

CRediT authorship contribution statement

Erikson J. de Aguiar: Conceptualization, Methodology, Formal analysis, Writing – original draft preparation, Writing – review & editing. **Alyson J. dos Santos:** Conceptualization, Methodology, Formal analysis. **Rodolfo I. Meneguette:** Conceptualization, Writing – original draft preparation, Writing – review & editing. **Robson E. De Grande:** Conceptualization, Writing – original draft preparation, Writing – review & editing. **Jó Ueyama:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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